



**NITI Aayog**

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SECTOR AT A GLANCE

# International Health Cooperation

Health Sector in India

DRAFT

Sector at a Glance

# **International Health Cooperation (IHC)**

Health Sector in India

The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

### **Suggested Citation:**

NITI Aayog (2023). International Health Cooperation. NITI Aayog Sector-at-a-Glance Series – Health in India, 12, <link>

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# Overview

As Paul Russel has very aptly remarked – *“Nothing on earth is more international than disease and disease has no political or geographical boundaries. Disease in any part of the world is a constant threat to other parts.”*<sup>1</sup>

India has actively participated in International Health Cooperation, cooperating with numerous foreign organisations and United Nations agencies to give technical support for many of the country's Health and Family Welfare programmes.<sup>2</sup>

India's bilateral and multilateral collaborations are facilitated by the International Health Division which helps in engagement with global organisations like the WHO (World Health Organization), UN (UNICEF (United Nations Children's Fund), UNDP (United Nations Development Programme), UN Fund for Population Activities, FAO (Food and Agriculture Organization), ILO (International Labour Organization), World Bank), USAID (United States Agency for International Development), The Colombo Plan, SIDA (Swedish International Development Cooperation Agency), DANIDA (Danish International Development Agency), Rockefeller Foundation, Ford Foundation, CARE (Cooperative for Assistance and Relief Everywhere), International Red Cross, Indian Red Cross.



## 1. International Health Cooperation

Global Health Cooperation began in the mid-19th century, shifting from national to global governance due to concerns regarding infectious diseases. The first **International Sanitary Conference in Paris in 1851** marked the start of international health diplomacy and was focused on cholera, plague, and yellow fever. Over the next century, cooperation on infectious diseases evolved.<sup>3</sup>

The **International Health Regulations (IHR)** is serving as the principal framework for global health security since **2007**. Central to the IHR is the declaration of Public Health Emergencies of International Concern (PHEIC).<sup>4</sup>

A **Public Health Emergency of International Concern (PHEIC)** is an official proclamation by the World Health Organization (WHO) indicating an extraordinary event that poses a public health risk to other nations through the international spread of disease and may necessitate a coordinated international response. Such declarations are made when a situation arises that is serious, sudden, unusual, or unexpected, with implications for public health extending beyond a single country's borders, and which may demand immediate international action.

Between 2007 and 2020, **six events** were **declared as PHEIC**: The 2009 H1N1 influenza pandemic, the Ebola outbreak (West Africa 2013–2015), the Ebola outbreak (Democratic Republic of Congo 2018–2020), ongoing poliomyelitis concerns since 2014, the Zika outbreak in 2016, and the ongoing COVID-19 pandemic since 2020. The COVID-19 pandemic has highlighted the vital need for national, regional, and international collaboration during crises, even though it's not the first or most severe global pandemic<sup>5</sup>

1 K Park. (2017). *Park's textbook of preventive and social medicine*.

2 Department of Health & Family Welfare. (2018). *Annual Report 2017-18*.

3 Fidler, D. (2001). *The globalization of public health: the first 100 years of international health diplomacy*.

4 Annelies Wilder-Smith, & Osman, S. (2020). *Public health emergencies of international concern: a historic overview*.

## Need for International Health Cooperation:



### Sharing knowledge and know-how

Cooperation among countries and horizontal partnerships optimize existing health capacities and encourage the sharing of knowledge and know-how between partners.<sup>5</sup> This can help countries to learn from each other's successes and failures, and to develop more effective health policies and programmes.



### Combining resources

Partnerships help organisations like the World Health Organization (WHO) to deliver life-saving vaccines and supplies, raise funds for global health targets, and establish consensus among leading scientists. Combining resources can help to achieve economies of scale and to maximize the impact of health interventions.<sup>6</sup>



### Tackling pandemics

International cooperation and coordination are essential to tackling pandemics like COVID-19 in terms of both assistance with emergency medical supplies and medical expertise.<sup>5</sup> The COVID-19 pandemic has highlighted the need for global collaboration in addressing health crises.

## International Health Division<sup>7,8</sup>

In the age of globalisation, characterised by an increased international travel and global exchanges, diseases have the capacity to swiftly traverse borders. An outbreak in one country can profoundly affect the well-being and economic stability of other nations. Therefore, preventing the international spread of diseases from one country to another is a significant and pressing challenge in the 21st century. As India is a signatory to the International Health Regulations 2005, the enforcement of public health measures at international entry points, such as airports, ports, and land border crossings, is a mandatory requirement. In India, these health units under IHP are responsible for undertaking measures for surveillance and response health activities at Airports/Seaports/Land Border are known as APHOs (Airport Health Organizations) / PHOs (Port Health Organizations) and Land Border Health Units (LBHUs).

The International Health Division within the Directorate of Global Health Security holds the mandate for overseeing both the administrative and technical aspects of entities such as Airport/Port Health Organisations, Land Border Quarantine Centres, and Yellow Fever Quarantine Centres, among others.

## Key International Health Cooperation Stakeholders

India is a country with a population of over 1.3 billion people, diverse cultures, languages, religions, and socio-economic conditions. Some of the important health issues India faces are communicable and non-communicable diseases, malnutrition, maternal and child mortality, environmental pollution, and lack of access to quality health care. To address these issues, India has been collaborating with various international partners. These partnerships aim to provide technical and financial assistance, share best practices and innovations, strengthen health systems and capacities, and promote global health security and solidarity.

<sup>5</sup> Pan American Health Organization. (n.d.). *Cooperation Among Countries for Health Development*. Retrieved 1 September 2023.

<sup>6</sup> World Health Organization. (n.d.). *How we collaborate*. Retrieved 1 September 2023.

<sup>7</sup> Directorate General of Health Services. (n.d.). *Public Health (International Health)*.

<sup>8</sup> Directorate General of Health Services. (n.d.). *International Health Division*.

## 1. International Organisations



The World Health Organisation is a specialised, non-political, health agency of the United Nations. The objective of the WHO is “*the attainment by all peoples of the highest level of health that will permit them to lead a socially and economically productive life.*” WHO has been working with the Government of India to achieve its health sector goals and bring transformative upgrades to the health sector under **the WHO India Country Cooperation Strategy (CCS) 2019- 2023**.

This CCS not only extends the groundwork that WHO has been actively involved in for the past several years but also broadens its scope to confront intricate issues, including but not limited to the prevention of **non-communicable diseases (NCDs), the management of antimicrobial resistance (AMR), the mitigation of air pollution, and the provision of care for mental health disorders.**<sup>9</sup>



**The United Nations Development Programme (UNDP)** is a UN agency that leads international efforts to eradicate poverty and achieve sustainable human development, an approach to economic growth that emphasises improving the quality of life of all citizens while conserving the environment. In India, UNDP supports the government's vision of faster and more inclusive growth, focusing on areas such as poverty reduction, democratic governance, energy and environment, disaster resilience, and women's empowerment.<sup>10</sup>



**UNICEF** is a UN agency that provides humanitarian and developmental assistance to children and mothers in developing countries. It was established in 1946 and its focus areas include child survival and development, education, child protection, social policy, and emergency response.

UNICEF collaborates with the Ministry of Health and Family Welfare, Women and Child Development, NITI Aayog, and state governments to support health sector planning, budgeting, policy formulation, capacity building, monitoring, and demand generation.<sup>12</sup>



The Joint United Nations Programme on HIV and AIDS (**UNAIDS**) is an innovative joint venture of 11 UN organisations that brings together the efforts and resources of different stakeholders to unite the world against AIDS. In India, UNAIDS works with the government, civil society, private sector and communities to provide strategic guidance, advocacy and technical support to accelerate the national AIDS response.<sup>13</sup>



The **World Bank** is a provider of financial and technical assistance to individual countries around the globe. The bank considers itself a unique financial institution that sets up partnerships to reduce poverty and support economic development. In India, the World Bank supports the country's development agenda through lending, knowledge products, policy dialogue and capacity building. Some of the key areas of engagement include social protection, health, education, rural development, urban development, energy, transport and climate change.<sup>14</sup>

9 WHO. (2019). *The WHO India Country Cooperation Strategy 2019–2023: A Time of Transition*.

10 UNDP. (n.d.). *About us*. Retrieved 4 September 2023.

11 UNDP. (n.d.). *Global Fund Partnership*. Retrieved 4 September 2023.

12 UNICEF India (n.d.). *What we do*. Retrieved 4 September 2023.

13 UNAIDS. (n.d.). *About UNAIDS*. Retrieved 4 September 2023.

14 World Bank. (2021). *75 Years of World Bank in India*. Retrieved 4 September 2023.



The **Asian Development Bank (ADB)** is a regional development bank that aims to promote social and economic development in Asia and the Pacific. ADB assists its members and partners by providing loans, grants, technical assistance and equity investments. In India, ADB supports the country's inclusive and sustainable growth through investments in infrastructure, skills development, urban services, rural connectivity, agriculture and natural resources management.<sup>15</sup>



The **International Labour Organization (ILO)** is a UN agency whose mandate is to advance social and economic justice by setting international labour standards. The ILO brings together governments, employers and workers to jointly shape policies and programmes promoting decent work for all women and men<sup>16</sup>. In India, the ILO works with the government and social partners to promote rights at work, enhance social protection, improve working conditions, create employment opportunities and foster social dialogue.<sup>16</sup>



The **United Nations Population Fund (UNFPA)** is a UN agency that aims to improve reproductive and maternal health worldwide. UNFPA's goal is to ensure sexual and reproductive rights and choices for all and make sure that people have regular access to enough high-quality food and lead active, healthy lives. In India, UNFPA supports the government's efforts to provide universal access to reproductive health services, end preventable maternal deaths, address gender-based violence and harmful practices, empower young people and advance gender equality.<sup>17</sup>



The **Association of Southeast Asian Nations (ASEAN)** is a political and economic union of 10 states in Southeast Asia, founded in 1967, that aims to promote regional cooperation, peace and stability, and socio-cultural development. In India, ASEAN is a key partner for trade, investment, connectivity, security and cultural exchange. India became a sectoral dialogue partner of ASEAN in 1992 and a full dialogue partner in 1996. India and ASEAN have established a strategic partnership in 2012 and have adopted a plan of action for 2021-2025 to enhance cooperation in various areas.<sup>18</sup>



The **Food and Agriculture Organization (FAO)** is a specialized agency of the UN that leads international efforts to defeat hunger. FAO's goal is to achieve food security for all and ensure that people have access to sufficient safe and nutritious food that meets their dietary needs and food preferences for an active and healthy life. In India, FAO collaborates with the government and other stakeholders to enhance agricultural productivity, improve food quality and safety, promote sustainable natural resource management, strengthen resilience to disasters and climate change, and foster rural livelihoods.<sup>19</sup>

<sup>15</sup> ADB. (n.d.). *Who we are*. Retrieved 4 September 2023.

<sup>16</sup> International Labour Organization. (n.d.). *About the ILO*. Retrieved 4 September 2023.

<sup>17</sup> UNFPA. (n.d.). *About us*. Retrieved 4 September 2023.

<sup>18</sup> ASEAN. (n.d.). *About ASEAN*. Retrieved 4 September 2023.

<sup>19</sup> FAO. (n.d.). *FAO in India*. Retrieved 4 September 2023.

## 2. Foreign Government Agencies



**USAID** is the lead U.S. government agency that works to end extreme global poverty and enable resilient, democratic societies to realise their potential. It was established in 1961 and its focus areas include global health, food security, democracy and governance, education, gender equality, environment and energy, and humanitarian assistance. USAID collaborates with the Government of India, the private sector, and civil society to improve the reach, affordability, and quality of health services for poor and vulnerable households, building on decades of USAID support for quality and accessible health care in India, including maternal and child mortality, polio, HIV, and tuberculosis.<sup>20</sup>

## 3. Public Private Partnerships



**Gavi** is a public-private partnership that aims to increase access to immunisation in poor countries. It was launched in 2000 and its focus areas include strengthening health systems, accelerating vaccine introduction, improving vaccine supply and sustainability, and shaping vaccine markets. Gavi has formed a strategic partnership with the Government of India to identify and vaccinate children who have never received a routine vaccine, strengthened its existing health systems, and assisted the country in incorporating the human papillomavirus (HPV) vaccine and typhoid conjugate vaccine (TCV) into the national routine immunisation schedule.<sup>21</sup>



**The Global Fund** is an international financing organisation that invests in programmes to fight HIV/AIDS, tuberculosis (TB) and malaria in more than 100 countries. The Global Fund mobilises resources from governments, the private sector and foundations to support countries in their response to these diseases. In India, the Global Fund has provided over USD 2 billion since 2002 to support programmes that have saved more than 1.3 million lives and prevented millions of new infections from HIV/AIDS, TB and malaria.<sup>22</sup>

## 4. Private International NGO's



**Bill and Melinda Gates foundation** is a private foundation that aims to improve the quality of life for people around the world. It was founded in 2000 by Bill and Melinda Gates and its focus areas include global health, global development, global policy and advocacy, and US education. It has an office in New Delhi and works with the Government of India and other partners to address key challenges in health, sanitation, agriculture, digital financial inclusion, and gender equality. Some of its initiatives include supporting the National Health Mission, the Swachh Bharat Mission, the National Nutrition Mission, and the Bharat Inclusion Initiative.<sup>23</sup>



**The Clinton Health Access Initiative (CHAI)** is a global health organisation that works to save lives by expanding access to essential medicines and health services for people living in low- and middle-income countries. CHAI partners with governments, donors, civil society and private sector actors to strengthen health systems and improve health outcomes. In India, CHAI works with the government and other partners to address key health challenges such as HIV/AIDS, TB, malaria, hepatitis C, cancer, diabetes and maternal and child health.<sup>24</sup>

<sup>20</sup> USAID. (n.d.). *India*. Retrieved 5 September 2023.

<sup>21</sup> Gavi. (n.d.). *About our Alliance*. Retrieved 4 September 2023.

<sup>22</sup> The Global Fund. (n.d.). *About the Global Fund*. Retrieved 4 September 2023.

<sup>23</sup> Bill & Melinda Gates Foundation. (n.d.). *India*. Retrieved 4 September 2023.

<sup>24</sup> Clinton Health Access Initiative. (n.d.). *About us*. Retrieved 4 September 2023.



**PATH** is a global health innovation organisation that works to accelerate health equity by bringing together public institutions, businesses, social enterprises, and investors to solve the world's most pressing health challenges. PATH focuses on areas such as vaccines, drugs, diagnostics, devices, and system and service innovations. In India, PATH works with the government and other partners to improve access to quality health care, strengthen health systems, and advance innovations in areas such as immunisation, digital health, nutrition, tuberculosis, maternal and child health, and COVID-19 response.<sup>25</sup>



**Jhpiego** is a global health non-profit organisation that works to prevent the needless deaths of women and families in low- and middle-income countries. Jhpiego designs innovative, effective and low-cost health care solutions to ensure a level of care for women and their families. In India, Jhpiego works with the government and other partners to improve the quality of maternal and newborn health services, prevent cervical cancer, expand family planning options, prevent and treat HIV/AIDS, and strengthen health systems.<sup>26</sup>



**John Snow, Inc. (JSI)** is a public health management consulting and research organisation that works to improve the health of individuals and communities in the US and around the world. JSI provides technical assistance, training, research, evaluation, logistics support, and information systems development for a range of health issues. In India, JSI works with the government and other partners to improve maternal and child health, immunisation, family planning, nutrition, HIV/AIDS prevention and care, tuberculosis control, and health systems strengthening.<sup>27</sup>

## Key Collaborations



### 1. G20<sup>28</sup>

The following three thematic priorities were identified with respect to health sector for the G20 meeting in 2023:

#### Priority I

Health Emergencies  
Prevention, Preparedness,  
and Response (with Focus  
on One Health and  
AMR).

#### Priority II

Strengthening Cooperation  
in Pharmaceutical Sector  
with focus on Access and  
Availability to safe, effective,  
quality and Affordable  
Medical Countermeasures  
(Vaccines, Therapeutics and  
Diagnostics).

#### Priority III

Digital Health Innovations  
and Solutions to Aid  
Universal Health Coverage  
and Improve Healthcare  
Service Delivery.

<sup>25</sup> PATH. (n.d.). *About us*. Retrieved 4 October 2023.

<sup>26</sup> Jhpiego. (n.d.). *About*. Retrieved 4 October 2023.

<sup>27</sup> JSI. (n.d.). *About us*. Retrieved 4 October 2023.

<sup>28</sup> G20. (n.d.). *G20 Health Ministers' Meeting*.

India's G20 priorities centre on reformed multilateralism, including creating a global R&D network that fosters innovation and accelerates global R&D network to improve pandemic preparedness and response by facilitating access to life-saving medical countermeasures.

## 2. World Health Organization<sup>29</sup>

WHO is one of the primary United Nations agencies that collaborates with the Ministry of Health and Family Welfare of the Government of India to provide technical skills and globally applicable high-quality norms and standards in a variety of areas, including:

- Noncommunicable diseases (NCDs) action plan roll-out
- Environmental health, including air pollution
- Mental health promotion and suicide prevention
- Nutrition and food safety
- Road safety
- Tobacco control
- Integration of NCD and environmental risk factors in the digital health information platform



## 3. BRICS<sup>30</sup>

The BRIC (Brazil, Russia, India, and China) leaders first met in St. Petersburg, Russia, in July 2006 and formalised as BRIC during the 1st BRIC Foreign Ministers' Meeting in New York in September 2006. BRIC became BRICS with the addition of South Africa in September 2010. BRICS nations have emerged as significant players in international health cooperation. Individually, each country has a track record of promoting development cooperation, shifting their role from mere donors to crucial "development partners." Brazil, China, and India in particular, stand out for their leadership in manufacturing affordable medicines and vaccines.

The collaboration among BRICS countries in the field of health is gaining traction, exemplified by the annual meetings of their health ministers. This collaborative effort serves as a valuable platform for sharing experiences and jointly addressing critical public health issues, including neglected tropical diseases. Leveraging their respective capacities and unique strengths, inter-BRICS cooperation holds the potential to instigate global change and positively impact public health not only within BRICS but also worldwide.

During the BRICS 2022 summit following resolutions were signed to boost health cooperation: The BRICS leaders reaffirmed their support for the WHO and other multilateral initiatives to combat COVID-19 and ensure equitable access to vaccines and health products. They agreed to enhance cooperation on public health emergencies, vaccine research and development, traditional medicine, digital health and TB control. They also called for strengthening local production of vaccines and boosters in developing countries and improving global health surveillance.<sup>31</sup>

<sup>29</sup> MoHFW. (2023). *Annual Report 2022-23*.

<sup>30</sup> Acharya, S., Barber, S.-L., López-Acuña, D., Natela Menabde, Migliorini, L., Molina, J., Bernhard Schwartländer, & Zurn, P. (2014). *BRICS and global health. Bulletin of the World Health Organization*.

<sup>31</sup> Ministry of External Affairs. (2022). *XIV BRICS Summit Beijing Declaration*.



#### 4. Conferences and Meetings<sup>29</sup>

- In 2022, the Ministry of Health and Family Welfare participated in 59 meetings with foreign organisations.
- In 2022, 38 applications for grants related to international conferences were processed.



#### 5. Airport/ Port Health<sup>29</sup>

The Indian government signed many memorandums of understanding with other governments to reduce the spread of viral diseases such as COVID and yellow fever.

## Global Collaborations for Pandemics

### 1. Preparedness and Resilience for Emerging Threats Initiative (PRET)

The World Health Organization (WHO) launched an initiative called the Preparedness and Resilience for Emerging Threats Initiative, or PRET, on 26 April, 2023 to bolster global readiness for future pandemics.<sup>32</sup> This innovative approach acknowledges that a unified set of systems, capabilities, knowledge, and tools can be efficiently adapted for different categories of pathogens, based on their modes of transmission, such as respiratory, vector-borne, or foodborne.<sup>33</sup>

PRET incorporates the latest tools and strategies developed during the COVID-19 pandemic and other recent public health crises, prioritising principles of fairness, inclusivity, and coherence. It acts as a platform for fostering collaboration among national, regional, and global stakeholders to bolster preparedness.<sup>32</sup>

#### Primary Objectives<sup>32</sup>



Facilitating increased collaboration among a wide range of stakeholders.



Advocating for an organised and equitable allocation of resources that enables coordinated efforts among countries and stakeholders.



Ensuring the transparent implementation of priorities established by national authorities, while aligning with regional and global directives.

### 2. UK-India COVID-19 Partnership Initiative<sup>34</sup>



This initiative offers funding for collaborative research projects aimed at addressing the public health challenges posed by the COVID-19 outbreak in both the United Kingdom and India. United Kingdom Research and Innovation has allocated up to GBP 4 million to support the UK segment of these collaborative projects, with an equivalent amount matched by the Department of Biotechnology to

<sup>32</sup> World Health Organization. (2023). *WHO launches new initiative to improve pandemic preparedness*. Retrieved 12 September 2023

<sup>33</sup> World Health Organization. (n.d.). *Preparedness and Resilience for Emerging Threats (PRET)*. Retrieved 12 September 2023

<sup>34</sup> Department of Biotechnology. (n.d.). *UK-India Covid-19 Partnership Initiative*.

support the Indian components. The primary goal is to provide research funding for collaborations that demonstrate international competitiveness and innovation, involving researchers from both India and the UK.



### Aim of the initiative

Through the UK-India trade route, the funders' aims for the COVID-19 Partnership Initiative are as follows:

1. Support world-leading collaborative research teams focusing on lowering the severity of COVID-19 among South Asian communities in the UK and India.
2. Strengthen current collaborations and forge new ones between the United Kingdom and India in order to combat COVID-19.
3. Strengthen the strategic partnership between the United Kingdom and India.



### Objective of the initiative

To provide research funding to support globally competitive and ground-breaking collaborations between researchers from India and the United Kingdom.

## 3. Gavi, The Vaccine Alliance<sup>35</sup>



Established in 2000, Gavi, the Vaccine Alliance, represents a cooperative endeavour involving both the public and private sectors with a shared commitment to immunising approximately 50% of the world's children against some of the deadliest diseases.

Additionally, Gavi plays a vital role in bolstering worldwide healthcare preparedness by extending assistance to healthcare systems and funding global vaccine reserves for conditions such as Ebola, cholera, meningococcal infections, and yellow fever.

### Mission:



To preserve lives and safeguard public health by promoting fair and long-lasting access to vaccines. We gauge our progress in achieving this mission through six "mission indicators," which serve as benchmarks for our objectives during the 2020–2025 period.

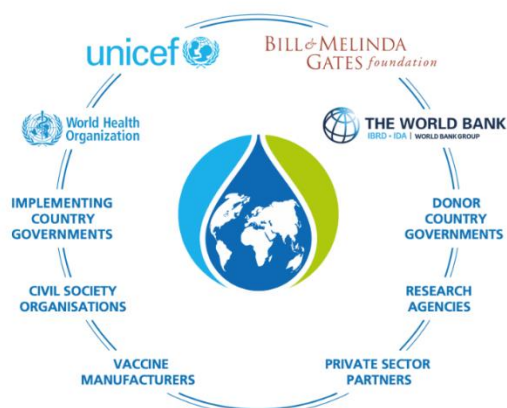


### Mission Indicators:

- Under-five mortality rate
- Future deaths averted with Gavi support
- Future disability-adjusted life years averted
- Reduction in the number of zero-dose children
- Unique children immunised through routine immunisation with Gavi support
- Economic benefits generated through Gavi-supported immunisations

<sup>35</sup> Gavi The Vaccine Alliance. (2021). *Annual Progress Report 2021*.

## Major Collaborators with Gavi, the Vaccine Alliance



Gavi's impact leverages the strengths of its core partners, the World Health Organisation, UNICEF, the World Bank, and the Bill and Melinda Gates Foundation, and plays a critical role in strengthening primary health care (PHC), bringing the world closer to the Sustainable Development Goal (SDG) of Universal Health Coverage (UHC), and ensuring that no one falls behind.

Gavi also collaborates with donors, such as sovereign governments, private sector foundations, and corporate partners; non-governmental organisations (NGOs), advocacy groups, professional and community associations, faith-based organisations, and academia; vaccine manufacturers, including those in emerging markets; research and technical health institutes; and implementing country governments.

### Gavi's contribution to International Health<sup>35</sup>



**> 981 Mn**

children vaccinated through routine programmes



**> 1.40 Bn**

vaccinations through vaccination campaigns



**>16.2 Mn**

future deaths prevented



**561**

vaccine introductions and campaigns



**>USD 185.3 Bn**

in economic benefits generated



**USD 1.3 Bn**

in co-financing contributions from Gavi-supported countries since 2008

## 4. India's Contribution to International Vaccine and Drug Needs During COVID-19 Pandemic

The Indian pharmaceutical industry has also played a crucial role in addressing the challenges posed by the COVID-19 pandemic. This sector actively collaborated with the government and academic institutions to swiftly develop and refine manufacturing processes, ensuring a consistent supply of essential medicines for the treatment of COVID-19, including Remdesivir, Ivermectin, Hydroxychloroquine, Dexamethasone, Tocilizumab, and Favipiravir.<sup>36</sup>

<sup>36</sup> Department of Pharmaceuticals. (2022). *Annual Report 2021-22*.

Throughout the pandemic, India's pharmaceutical exports helped approximately 96 countries with vaccines, more than 120 countries with Hydroxychloroquine (HCQ) and 20 countries for paracetamol offering much-needed relief on a global scale.

By 30th November 2022, a total of **28.1 crore** doses of Made-in-India COVID-19 vaccines were dispatched to **99** countries along with UN Peacekeepers and UN Health Workers as part of the **Vaccine Maitri** initiative.<sup>37</sup>



## 2. International Research Cooperation



**icmr** भारतीय आयुर्विज्ञान  
अनुसंधान परिषद्  
INDIAN COUNCIL OF  
MEDICAL RESEARCH

International research cooperation in health is of great importance. It maximizes the resources and skills available for dealing with global health issues, builds capacity and endeavours to ensure the research reflects real needs of the populations. Medical research and technology development can be benefited greatly through international cooperation and collaboration. Common goals can be effectively addressed by pooling both material and intellectual resources.

The Indian Council of Medical Research (ICMR), headquartered in New Delhi, is the preeminent institution in India responsible for developing, coordinating, and fostering biomedical research. It stands as one of the world's oldest organisations dedicated to medical research. The International Health Division (IHD) in ICMR co-ordinates international biomedical research between India and other countries and agencies. The IHD also aims to enhance India's performance in collaborative research.

Throughout its history, the ICMR has consistently strived to meet two crucial objectives. Firstly, it has focused on adapting to the ever-evolving landscape of scientific progress in biomedical research. Secondly, it has dedicated itself to finding practical solutions to address the healthcare challenges faced by the nation.

### International Health Measures by ICMR<sup>38</sup>

#### A. Health Ministry's Screening Committee (HMSC)

The Department of Health Research/ICMR, under the Ministry of Health and Family Welfare, Government of India, has undergone a recent reconstitution of its Screening Committee, known as HMSC, in accordance with the directive issued by the Cabinet Secretary. The HMSC is presided over by the Secretary of the Department of Health Research (DHR), MOH&FW, GoI, while the Vice-Chairperson of the Committee holds the position of Secretary in the Department of Biotechnology (DBT).

The Department of Health Research (DHR/ICMR) serves as the administrative body for HMSC and plays a crucial role in overseeing the technical evaluation of international collaborative proposals. This evaluation is a mandatory prerequisite before presenting the proposals to the Committee.

#### How does HMSC operate?

- The newly reconstituted HMSC is tasked with the assessment and approval of proposals that encompass international collaboration and funding for health research. This includes studies encompassing human, animal, plant, and environmental aspects and involves non-governmental organisations.
- The Committee is scheduled to convene regularly, with meetings occurring at least once every two months, to review and approve projects submitted for consideration by the HMSC.

<sup>37</sup> MEA. (2022). *Annual Report 2022*.

<sup>38</sup> Ministry of Health and Family Welfare. (n.d.). *Health Ministry's Screening Committee (HMSC) Guidelines*. Retrieved 12 September 2023

## B. The Biological Research Regulatory Approval Portal (BioRRAP)

- BioRRAP, overseen by the Department of Biotechnology (DBT), Government of India, has been established to provide researchers with regulatory information and streamline the approval process.
- This innovative gateway is designed to enhance accessibility to science and scientific research while simplifying the process of establishing businesses in this sector.
- Going forward, the Government of India has mandated that all proposals submitted for HMSC consideration must first register on BioRRAP to obtain a unique BioRRAP ID. Subsequently, these proposals can be submitted on the HMSC portal with a valid BioRRAP ID.

## Other Mandates

- ICMR scientists are obligated to seek HMSC clearance for health research projects in the international domain that involve foreign funding and/or collaboration, including those under bilateral programmes with other Indian funding agencies.
- In addition to this requirement, proposals put forth by ICMR Institutes and Centres must also include the recommendations from their respective institute or centre's Scientific Advisory Committee (SAC).

### India Africa Health Sciences Platform (IAHSP)<sup>39</sup>



#### Pillars of India-Africa Partnership

Social

Financial

Political

The collaboration between India and Africa is touted as partnership of equals

The India Africa Health Sciences Platform (IAHSP) is a collaborative initiative between India and Africa to promote health research, capacity building and development. The IAHSP was established by the Indian Council of Medical Research (ICMR) and the African Union (AU) in 2019, following the vision of the India-Africa Forum Summit (IAFS) III in 2015. The IAHSP aims to address diseases of common concern in both regions through indigenous development of affordable drugs, diagnostics, and vaccines, and also enable knowledge sharing and capacity strengthening. The IAHSP offers training courses for health practitioners and researchers from various African countries in different ICMR institutes in India. The IAHSP is a partnership of equals, based on mutual respect and benefit, and reflects the strong ties between India

<sup>39</sup> Indian Council for Medical Research. (n.d.). *India Africa Health Sciences Platform*.

and Africa.

### Common Concerns



Health



Healthcare  
Delivery



Poverty



Nutrition



Sanitation



Infrastructur  
e

### India Africa Health Sciences Meet (IAHSM)

- The India Africa Health Sciences Meet (IAHSM) is a **collaborative initiative between India and Africa** to promote and strengthen health sciences research and innovation in both regions.
- The IAHSM was **launched in 2016 by the Indian Council of Medical Research (ICMR) and the African Union (AU)** with the participation of various stakeholders from both sides.
- The IAHSM aims to address common health challenges such as **HIV/AIDS, tuberculosis, and cancer** by developing affordable drugs, diagnostics, and vaccines locally, while also fostering knowledge exchange and capacity-building.
- The IAHSM also includes a research grant scheme and a capacity building scheme for health practitioners and researchers from India and Africa.
- The IAHSM is a testament to the strong partnership and mutual cooperation between India and Africa in the field of public health.





### 3. Antimicrobial Resistance (AMR)<sup>40</sup>



#### AMR: A Threat

Antimicrobial resistance (AMR) is a global public health threat that requires urgent multisectoral action. It makes infections harder to treat and increases the risk of disease spread, along with severe illness and death.

#### Drivers of AMR:<sup>41</sup>



**Improper Use of Antimicrobials** – Misuse and overuse of antimicrobial drugs, in both human and animal healthcare by self-medicating, using antibiotics when not needed, and not following the prescription leading to incomplete treatment regimen, accelerates the development of resistant bacteria.



**Limited Access to Clean Water, Sanitation and Hygiene (WASH) for Both Humans and Animals** – Limited access to clean water, proper sanitation facilities, and good hygiene practices increases the risk of infections and the need for antimicrobial treatment.



**Infection and Disease Prevention and Control** – Improper infection and disease prevention and control contribute to the transmission of drug-resistant infections. Inadequate hygiene practices, insufficient sterilisation of medical equipment, and improper use of infection control protocols create opportunities for resistant bacteria to thrive and spread.



**Access to Quality, Affordable Medicines, Vaccines, and Diagnostics** – The limitation of not being able to afford quality treatment, hampers effective control of infections which can lead to inadequate or inappropriate use of antimicrobials and promote the development of resistance.



**Awareness and Knowledge** – Insufficient awareness and knowledge about the proper use of antibiotics, the risks of AMR, and the importance of infection prevention contribute to inappropriate antibiotic use and the spread of resistance.



**Enforcement of Legislation** – Inadequate regulation of antibiotic sales, surveillance of antimicrobial use, and a lack of penalties for non-compliance with guidelines.

<sup>40</sup> WHO. (n.d.). *Antimicrobial Resistance*. Retrieved 12 September 2023.  
<sup>41</sup> World Health Organization. (2019). *AWaRe Classification Database*.

## Global Action Plan on Antimicrobial Resistance (GAP)<sup>40</sup>

The Global Action Plan on Antimicrobial Resistance (GAP) is a resolution adopted by the World Health Assembly in 2015 to address the growing problem of antimicrobial resistance (AMR). The GAP outlines five strategic objectives:

- To improve awareness and understanding of AMR through effective communication, education and training.
- To strengthen surveillance and research on AMR.
- To reduce the incidence of infection through preventive measures and infection control.
- To optimise the use of antimicrobial medicines in human and animal health.
- To enhance collaboration among different sectors and stakeholders using the “One Health” approach.

The creation of this global action plan on antimicrobial resistance, which was proposed by the World Health Assembly in resolution WHA 67.25 in May 2014, represents a global consensus that antimicrobial resistance poses a serious threat to human health.

The GAP also calls for countries to develop and implement national action plans (NAPs) on AMR, based on the GAP framework and aligned with their specific contexts and needs. India has already adopted The GAP and also urges for more investment in innovation and research to develop new medicines, diagnostics, vaccines and other interventions to combat AMR.

### Other Key Initiatives by WHO regarding AMR

- Tripartite Joint Secretariat on Antimicrobial Resistance
- World Antimicrobial Awareness Week (WAAW)
- The Global Antimicrobial Resistance and Use Surveillance System (GLASS)
- Global Antibiotic Research and Development Partnership (GARDP)

## International collaborations by ICMR to tackle AMR<sup>42</sup>

### ICMR-RCN collaboration



A Memorandum of Understanding (MoU) signed in 2014 to establish a health research relationship on human vaccines, infectious diseases and antimicrobial resistance (AMR) between ICMR and the Research Council of Norway (RCN). A workshop held in 2016 to discuss the challenges and opportunities of AMR research and collaboration, focusing on four themes: surveillance, stewardship, diagnosis and treatment.

### AMR Diagnostics Use Accelerator



A project by ICMR and FIND to stimulate studies on the effect of innovative clinical algorithms using existing diagnostic tools and behaviour change initiatives to improve antibiotic prescribing practices and management of febrile illnesses.

<sup>42</sup> Indian Council for Medical Research. (n.d.). *Collaborations*. Retrieved 8 September 2023.

## ICMR-AIIMS-CDC Global Health Security Agenda (GHSa) Collaborative Project



A project initiated in 2015 to strengthen, standardise and expand the capacity of the public health and healthcare systems in India to generate, apply and report accurate AMR data, implement infection control procedures, enhance antimicrobial stewardship practices and strengthen surveillance for healthcare associated infections (HAIs).

## ICMR-NIAID collaboration



A letter of Intent signed in 2015 to collaborate on research on AMR with National Institute of Allergy and Infectious Diseases (NIAID) using systems biology approach to understand the transmission dynamics and molecular mechanisms of drug resistant infections and develop improved therapeutics and diagnostics.

## ICMR-GARDP collaboration



An MoU signed with Global Antibiotic Research and Development Partnership (GARDP) to work towards development of new drugs to handle AMR crisis in India, with a goal to deliver 5 treatments by 2025 in four areas: serious bacterial infections, neonatal sepsis, sexually-transmitted infections and children's antibiotics.

## ICMR-BMBF collaboration



A collaboration with Federal Ministry of Education and Research (BMBF), Germany to strengthen and expand the basic research activities in AMR as part of global initiatives, focusing on key areas like mechanism of resistance, understanding pathogens, host responses, immunomodulation, vaccination and new antibiotic development and treatment strategies.

## ICMR-Pfizer collaboration



A collaboration with Pfizer Ltd. to institute a 'Centre to Combat Antimicrobial Resistance' in New Delhi, as part of the MoU signed in 2017. The joint initiative will implement a series of comprehensive interventions, ranging from AMR stewardship programmes for nursing homes to scaling up of the ongoing AMR surveillance network and creating awareness around responsible use of antibiotics.

## ICMR-ICAR-FAO collaboration<sup>43</sup>



A collaboration with Indian Council of Agricultural Research (ICAR) and Food and Agricultural Organisation (FAO) launched in 2017 to build capacity for integrated surveillance of AMR in food producing animals, food of animal origin and their environment and foodborne pathogens in humans. The project will engage with veterinary microbiologists, develop standard operating procedures for antimicrobial susceptibility testing for veterinary labs, identify capacities and gaps in selected labs and initiate

<sup>43</sup> FAO. (2017). *FAO-ICAR meeting on surveillance of AMR in human health and animal health sectors*. Retrieved 3 October 2023.



## 4. Medical and Wellness Tourism

- The Indian medical tourism, market is worth INR 0.4 lakh crore and is expected to grow to INR 1.06 lakh crore by 2026.<sup>45</sup>
- Medical tourism is now also popularly known as Medical Value Travel (MVT) because it encompasses both patients' healthcare-seeking behavior and the broader economic impact of such travel on the country that hosts them.<sup>46</sup>
- Over the last decade, India has built a reputation for providing high-quality medical care at low costs to medical tourists from all over the world, and it has now emerged as a global leader in the medical tourism sector.<sup>46</sup>
- The global wellness tourism market was worth INR 53,109 Lakh Crore in 2017, according to the Global Wellness Institute, with secondary wellness travellers accounting for 89 percent of trips and 86 percent of expenditure.<sup>47</sup>
- In terms of market size, the leading countries for wellness tourism are the United States, Germany, China, France, Japan, Austria, and India.<sup>47</sup>

### Key Components<sup>45</sup>

Medical & Wellness tourism refers to travel-related activities and the hosting of a foreign tourist who stays at least one night in the destination region to maintain, improve, or restore health through medical intervention.



#### Medical Treatment

Curative treatment may involve cardiac surgery, organ transplantation, hip and knee replacement, and other procedures.



#### Wellness and Rejuvenation

Offerings aimed towards rejuvenation or aesthetic purposes, such as cosmetic surgery, stress treatment, spas, etc.



#### Cures through Traditional System of Medicines

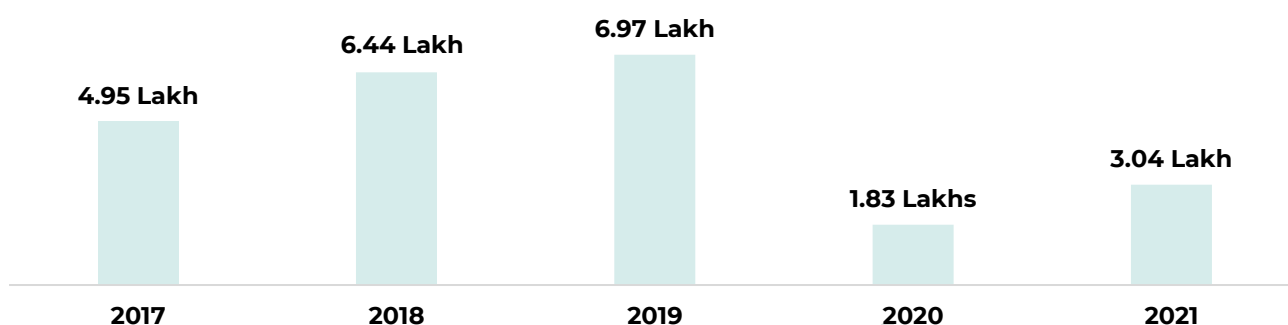
Access to traditional medical systems, such as India's AYUSH (Ayurveda, Yoga & Naturopathy, Unani, Siddha, and Homoeopathy).

<sup>44</sup> Ministry of Tourism. (2021). *Status of Medical Tourism in India*.

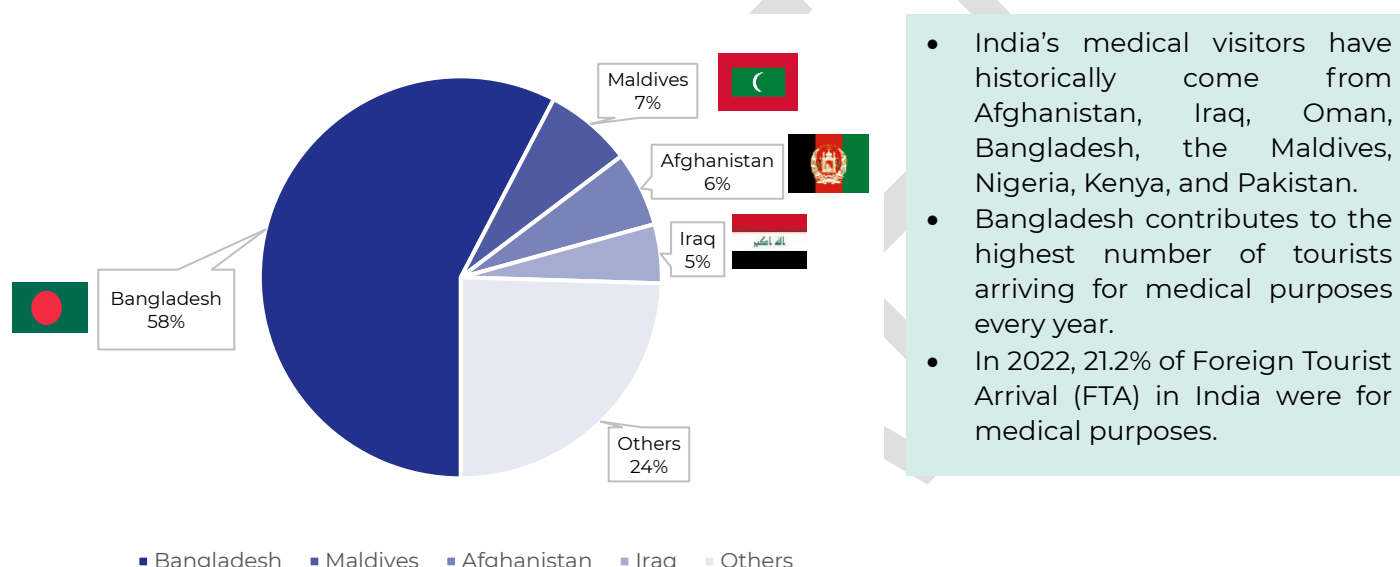
<sup>45</sup> Invest India. (n.d.). *Medical Value Travel*. Retrieved 12 September 2023.

<sup>46</sup> Ministry of Tourism. (2022). *National Strategy and Roadmap for Medical and Wellness Tourism*.

**Fig 1 - Number of Foreign Tourist Arrivals in India on Medical Visa<sup>47</sup>**

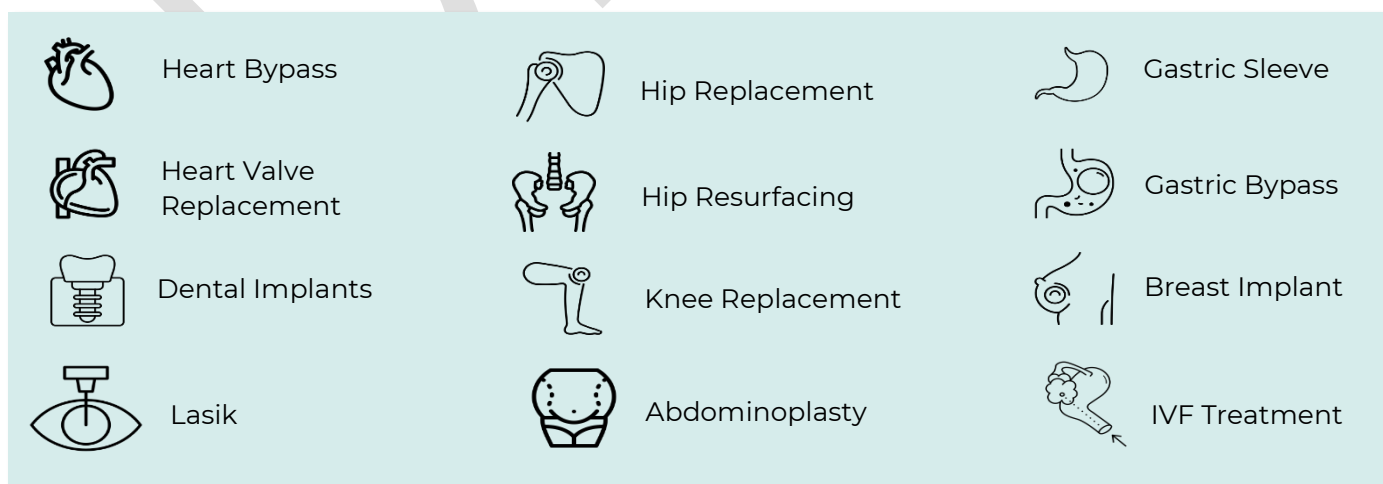


**Fig 2 - Percentage Distribution of Foreign Tourists' Arrivals from Leading Countries for Medical Purpose During 2022<sup>48</sup>**



### Medical Procedures in India cost lower than Major Destinations for Medical Value Travel<sup>45</sup>

India offers world class treatment at savings of approximately 65-90% compared to the USA. Some of the popular Medical procedures for MVT patients include:



<sup>47</sup> Ministry of Tourism. (2023). *Policy for emerging medical tourism*.

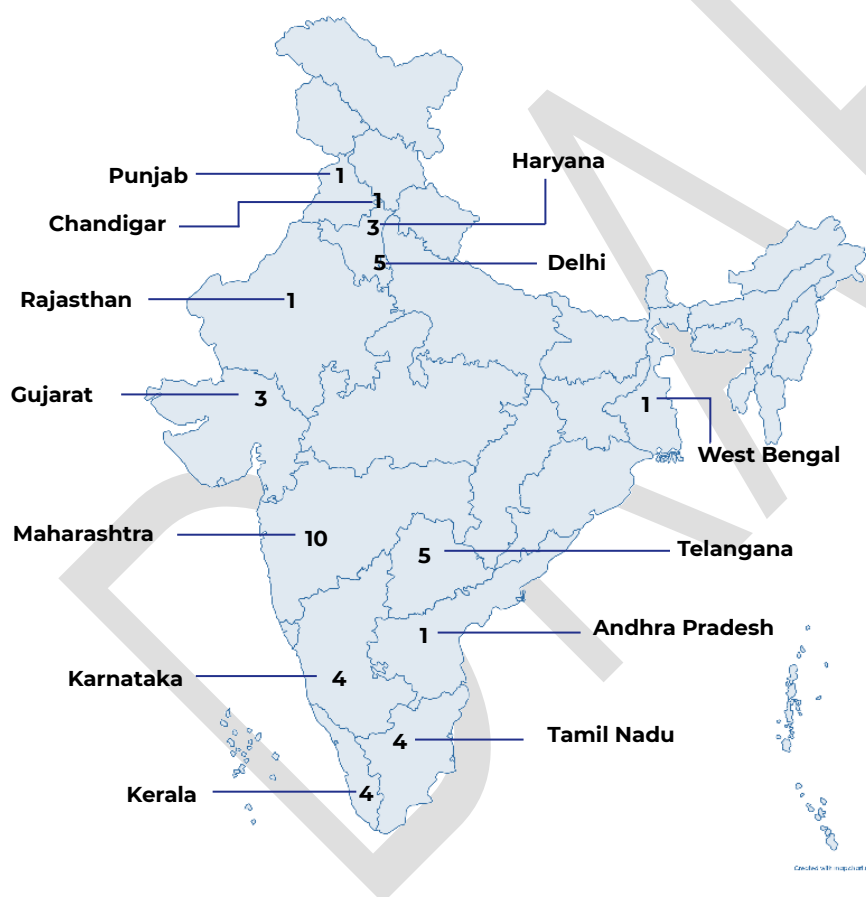
<sup>48</sup> Ministry of Tourism. (2022). *India Tourism Statistics 2022*.

**Table 1 - Prices of Common Medical Procedures Across Destinations of Medical Value Travel (in USD)<sup>49</sup>**

| Medical Procedure       | India | Thailand | Malaysia | Singapore | Turkey | South Korea |
|-------------------------|-------|----------|----------|-----------|--------|-------------|
| Heart Bypass            | 7,900 | 15,000   | 12,100   | 17,200    | 13,900 | 26,000      |
| Heart Valve Replacement | 9,500 | 17,200   | 13,500   | 16,900    | 17,200 | 39,990      |
| Hip Replacement         | 7,200 | 17,000   | 8,000    | 13,900    | 13,900 | 21,000      |
| Hip Resurfacing         | 9,700 | 13,500   | 12,500   | 16,350    | 10,100 | 19,500      |
| Knee Replacement        | 6,600 | 14,000   | 7,700    | 16,000    | 10,400 | 17,500      |

|          |             |           |
|----------|-------------|-----------|
|          |             |           |
| Low Cost | Medium Cost | High Cost |

**Fig 3 - State Wise Availability of Joint Commission International (JCI) Accredited Hospitals in India<sup>46</sup>**



- International Accreditations serve as an assurance to the international patients that the healthcare services provided by accredited hospitals meet global benchmarks.
- Southern and Western States of India have a greater concentration of Medical Value Travel service providers.
- India has around 37 Joint Commission International (a global health care accreditation body) accredited hospitals as well as 513 National Accreditation Board for Hospitals & Healthcare Providers (NABH) accredited hospitals.

<sup>49</sup> Niti Aayog. (2021). *Investment Opportunities in Healthcare Sector*.

# Opportunities and Challenges

## Opportunities



### India's Health Sector Gets a Boost from World Bank and USAID Loans

- The Government of India and the World Bank signed two INR 4,000 Crore complementary loans to assist and boost India's health sector development.<sup>50</sup>
- World Bank has committed to contribute INR 8,000 Crore to India's flagship Pradhan Mantri-Ayushman Bharat Health Infrastructure Mission (PM-ABHIM), to boost public healthcare infrastructure across the country.
- USAID is working with India's National Health Authority to develop a flexible financing mechanism to improve health outcomes in the country. The goal of this "blended financing facility" is to promote innovative market-based.<sup>51</sup>



### AYUSH – A unique wellness proposition

- India has a unique advantage to attract medical value travellers who seek a cure and wellness through AYUSH, which is a huge investment in wellness and alternative healing for the fast and hectic modern lifestyle.<sup>45</sup>



### Medical Value Travel becoming more accessible

- There has been a significant improvement in air connectivity from various source markets, opening up more options for Medical Value Travel.<sup>45</sup>



### Leadership in AMR research efforts

- India has already launched the BIRAC initiative to catalyse AMR research in India. International alliances like India playing a leadership role on the steering group of Global Health Security Agenda (GHSA) and joining the Global AMR Research and Development hub.<sup>52</sup>

## Challenges



### Skilled Manpower issues in wellness Industry<sup>45</sup>

- Inadequate human resources due to lack of standardised and accredited courses has resulted the wellness industry to have very limited pool of skilled professionals.



### Unorganized Medical Value Travel Sector<sup>45</sup>

- The Medical Value Travel facilitators are not well-organized or accredited which leads to many unethical agents taking advantage of the medical value traveller.
- There are still only a few JCI-accredited hospitals in India, and there is little international awareness of NABH accreditation.
- Unavailability of insurance policies at the international level make medical travel less attractive.

<sup>50</sup> World Bank. (2023). *World Bank Signs a \$1 Billion Program to Support India's Health Sector for Pandemic Preparedness*. Retrieved 12 September 2023

<sup>51</sup> USAID. (n.d.). *Transforming Development Through Innovation and Partnership*. Retrieved 12 September 2023.

<sup>52</sup> Indian Council of Medical Research. (n.d.). *India's Antimicrobial Resistance Surveillance & Research Initiative*. Retrieved 20 September 2023.



### Implementation of International Health Cooperation in a uniform manner

- India's diverse cultural beliefs and norms require customized approaches in healthcare delivery, which poses a challenge in implementing the standardized international approaches.
- India's varying socio-economic landscape leads to unequal access to healthcare, hindering the uniform application of international health initiatives.

## Cross Cutting Themes

### Climate



**Significance:** Climate change is a risk across borders and poses serious threats to the health and well-being of billions of people worldwide, especially the poor and vulnerable. Hence, International healthcare cooperation will play a vital role in working towards solutions. It can increase the risk of heat-related illnesses, waterborne and vector-borne diseases, malnutrition, mental health problems, and injuries from extreme weather events.

**Current Efforts:** India has participated in various global platforms and initiatives, such as the Paris Agreement, the Kigali Amendment, the WHO Framework Convention on Tobacco Control, and the Global Action Plan for Healthy Lives and Well-being for All. India has also collaborated with various countries and regions, such as the UK, the EU, the US, and Africa, on areas such as renewable energy, clean air, disaster risk reduction, and health system strengthening.

**Way Forward:** India needs to continue and enhance its international cooperation on climate change and health, by aligning its national policies and plans with the global goals and commitments, sharing its best practices and lessons learned with other countries, and leveraging its strengths and capacities in innovation, technology, and human resources.

### Education and Awareness



**Significance:** International health cooperation can enable the sharing of scientific information and data, the access to advanced technologies and infrastructure, the mobilisation of financial and human resources, and the creation of networks and platforms for joint research and innovation in priority areas of mutual interest and benefit.

**Current Efforts:** India has also engaged in international cooperation on education and awareness with various partners, such as WHO, UNESCO, UNICEF, USAI, UK, Germany, Japan, Australia, and Canada.

**Way Forward:** India also needs to strengthen the linkages between education and health sectors, such as by promoting health education in schools, enhancing health-related skills training, fostering research collaboration, and facilitating student and faculty mobility.

## Research and Education



**Significance:** Research and development (R&D) is vital for generating new knowledge, innovations, and solutions for addressing the health challenges faced by India and the world. R&D can contribute to improving health outcomes, quality of care, efficiency of service delivery, and competitiveness of the health sector.

**Current Efforts:** India has a vibrant R&D ecosystem that involves various stakeholders such as government agencies, academic institutions, private sector, civil society, and international partners. India has also participated in various international cooperation initiatives on R&D, such as the Coalition for Epidemic Preparedness Innovations (CEPI), the Global Alliance for Vaccines and Immunization (GAVI), the International Vaccine Institute (IVI), the Global Antibiotic Research and Development Partnership (GARDP), the International Centre for Genetic Engineering and Biotechnology (ICGEB), and the International Solar Alliance (ISA).

**Way Forward:** India needs to increase its R&D expenditure as a percentage of GDP, which is currently below the global average. India also needs to enhance the coordination, collaboration, and communication among different actors in the R&D system, as well as align R&D priorities with national health goals and global standards.

## Financial Aid



**Significance:** Financial aid is crucial for ensuring universal health coverage, reducing out-of-pocket expenditure, enhancing financial protection, and improving access to quality health care for all. Financial aid can also mobilise resources, leverage partnerships, and foster accountability in the health sector.

**Current Efforts:** India receives financial support from various international sources, such as bilateral and multilateral agencies, foundations, and private sector, for implementing health projects and programmes in areas such as maternal and child health, HIV/AIDS, tuberculosis, malaria, polio, and COVID-19. India also contributes to burning healthcare issues worldwide both financially and by leveraging its huge medical manpower.

**Way Forward:** India needs to expand the coverage, scope, and effectiveness of its financial aid schemes and programmes, as well as streamline their implementation and monitoring. India also needs to explore innovative sources and mechanisms of financing for health care, such as taxation, social insurance, public-private partnerships, and external assistance.

# Ministries and Departments

In Indian governance, policy convergence plays a critical role. It involves the coordination and collaboration of different ministries and departments to attain shared goals and objectives.

**Table 2 - Ministries and Departments**

| Ministry                                                                                                                          | Department                                                                                                                            | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ministry of Health and Family Welfare</b><br> | <b>Department of Health Research (DHR)</b><br>       | The Department of Health Research funds and monitors various health research projects and programmes, such as the Indian Council of Medical Research, the National Institute of Health and Family Welfare, the National Institute of Communicable Diseases, and the National Institute of Immunology. It also facilitates the exchange of scientific information and expertise with other countries and organisations on health research issues.                |
|                                                                                                                                   | <b>Indian Council of Medical Research (ICMR)</b><br> | The Indian Council of Medical Research is the apex body in India for the formulation, coordination and promotion of biomedical research. It conducts and supports research on various health issues, such as infectious diseases, chronic diseases, maternal and child health, nutrition, environmental health, and traditional medicine. It also collaborates with international agencies and organisations on health research projects and programmes.        |
|                                                                                                                                   | <b>International Health Division</b>                                                                                                  | International Health Division is responsible for preventing and responding to public health risks at international points of entry in India, following the International Health Regulations and other rules.                                                                                                                                                                                                                                                    |
| <b>Ministry of Science and Technology</b><br>  | <b>Department of Biotechnology (DBT)</b><br>       | The Department of Biotechnology is responsible for planning, promoting and coordinating biotechnology research and development in India. It supports various biotechnology initiatives, such as vaccine development, bioinformatics, genomics, proteomics, stem cell research, bioengineering, and nanobiotechnology. It also fosters international cooperation and partnerships in biotechnology through bilateral and multilateral agreements and programmes. |

| Other Important Ministries working on International Healthcare Cooperation                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <b>Ministry of External Affairs</b> | The Ministry of External Affairs is the primary agency for conducting India's foreign relations, including diplomatic, consular, and cultural affairs. The Ministry also oversees India's development cooperation with other countries, especially in the fields of health, education, agriculture, and energy. The Ministry also facilitates the participation of Indian experts and institutions in various international health forums and networks. |
|  <b>Ministry of Cooperation</b>      | The Ministry of Cooperation is a newly created ministry that aims to strengthen the cooperative movement in India and to promote cooperative values and principles. The Ministry also seeks to enhance India's engagement with other countries and organisations on cooperative issues, including health cooperatives that provide affordable and quality health care services to their members.                                                        |
|  <b>Ministry of Ayush</b>            | The Ministry of Ayush is dedicated to promoting and developing the traditional systems of medicine and health care in India, such as Ayurveda, Yoga, Naturopathy, Unani, Siddha, and Homeopathy. The Ministry also works to enhance the international recognition and acceptance of these systems, as well as to foster cooperation and collaboration with other countries and organisations in the field of traditional medicine.                      |

# Government Interventions

## Digital Health Innovation Network<sup>28</sup>

- In 2018, the G20 launched the Digital Health Innovation Network (DIN), a forum for G20 countries to share digital health expertise and best practices.
- In addition, the DIN has created a variety of tools, including a white paper on digital health and a toolbox for implementing digital health policy.

## The WHO India Country Cooperation Strategy 2019–2023 (CCS)<sup>9</sup>

- The WHO India Country Cooperation Strategy 2019-2023 (CCS) is a strategic framework for WHO to cooperate with the Government of India (GoI) to achieve health sector goals, improve population health, and bring about revolutionary reforms in the health sector.

## BRICS Health Ministers Meeting<sup>30</sup>

- The 12th Meeting of the BRICS Health Ministers marked a significant step forward in BRICS health cooperation.
- The gathering aided in strengthening the BRICS countries' partnership and identifying new areas for collaboration.

## Central Sector Scheme for Promotion of International Co-operation in AYUSH<sup>53</sup>

- The International Cooperation (IC) Scheme was established to address developing needs for international AYUSH promotion.
- Many countries, including Mauritius, Germany, Morocco, and Suriname, have expressed interest in building Ayurveda centres and institutions in the past five years.

## National Strategy for Medical Value Travel and Wellness Tourism Pillars<sup>46</sup>

- Develop a brand for India as a wellness destination
- Strengthen the ecosystem for medical and wellness tourism
- Enable digitalization by setting up Online Medical Value Travel (MVT) Portal
- Enhancement of accessibility for Medical Value Travel
- Promoting Wellness Tourism
- Governance and Institutional Framework

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<sup>53</sup> Ministry of Ayush. (2021). *Central Sector Scheme for Promotion of International Co-operation in AYUSH*.

## Other Initiatives<sup>46</sup>



The Ministry of Tourism developed a **National Strategy and Roadmap for Medical and Wellness Tourism** to create a strong framework and synergy across Central Government Ministries, State Governments, and the Private Sector to promote India as a Medical and Wellness Tourism destination.



### Heal in India

The Indian government has launched the 'Heal in India' campaign to promote medical and wellness tourism in the nation.



### Medical Visa and Medical Attendant Visa

India has introduced the Medical Visa and Medical Attendant Visa, with e-medical visas available in 150+ countries within 24-48 hours.

# Way Forward



## Bringing International Talents to India<sup>54</sup>

Establishment of centres of excellence focused on developing technologies like Artificial Intelligence (AI) and Machine Learning (ML), will help build credibility and health research institutes like NIPER (National Institute of Pharmaceutical Education and Research) need to attract international chairs to bring international talents.



## Partnerships to expedite R&D and access to life-saving medical countermeasures<sup>28</sup>

As an effort to ensure fair distribution and access to life-saving medical treatments against epidemic and endemic diseases, countries need to keep working together on R&D projects.



## Cooperation for collaboration on Digital Health and One Health<sup>55</sup>

India and France signed a Letter of Intent for Cooperation in the field of health and medicine, laying the groundwork for future collaboration in areas such as digital health, artificial intelligence for healthcare, medical waste treatment technologies, biotechnology, and the One Health concept, among others.



## Continued Growth of Rejuvenation Centres<sup>46</sup>

Yoga, Ayurveda, and Wellness is being promoted under the Ministry of Tourism's "Incredible India Campaign".

From yoga ashrams to spas and wellness centres offering holistic therapies, India offers a multitude of sites and locations to retreat, recharge, and rejuvenate.



## More Favourable Legislation<sup>45</sup>

The Government of India recognises medical value travel and wellness tourism as vital areas with the potential to accelerate India's growth and achieve the Aatmanirbhar Bharat goal.



## Government Collaborations<sup>45</sup>

In order to promote medical value travel, the government will look at corporate or institutional arrangements with Governments of countries that are looking to take care of the medical needs of their citizens.

<sup>54</sup> Department of Pharmaceuticals. (2021). *Draft policy to Catalyse Research & Development and Innovation in the Pharma-MedTech Sector*.  
<sup>55</sup> MEA. (2023). *Horizon 2047*. Retrieved 16 September 2023.

### **Strengthening India's leadership on AMR<sup>56</sup>**



Strengthening India's leadership through international collaborations, contributing to global efforts to contain AMR, while also emphasising national and state-level collaborations to facilitate coordination among disease control programmes and stakeholders for effective action against AMR at the ground level.

### **One Health Approach<sup>56</sup>**



The Department of Global Coordination and Partnership (GCP) for AMR collaborates with the Food and Agriculture Organization (FAO), the World Organisation for Animal Health (OIE), the United Nations (UN), and other entities, including civil society and the private sector, to lead and synchronise a global, multisectoral One Health response to AMR.

### **Global Health Diplomacy<sup>57</sup>**



India can leverage its position in global health diplomacy to advocate for more equitable health distribution, particularly for developing countries by continuing to play a pivotal role in shaping global health agendas, promoting South-South cooperation, showcasing leadership, etc. Recently, India's move on including African Union as a member of the G20, is an example of Global Health Diplomacy.

<sup>56</sup> National Centre for Disease Control. (2014). *National Action Plan, Antimicrobial Resistance*.  
<sup>57</sup> World Economic Forum

# Best Practices

## 1. “Leveraging Technology for Addressing the Third Delay\* in Maternal Health, Madhya Pradesh”<sup>58</sup>



### Background

Alliance for Saving Mother and Newborn or “ASMAN” initiative is driven by partners like Bill and Melinda Gates Foundation, Merck Sharp and Dohme for Mothers, Reliance Foundation, Tata Trusts and the USAID. Government of Madhya Pradesh served as the implementation partner for the initiative to tackle maternal and neonatal deaths caused by third delay.



### Steps taken by the government

As a part of the effort, modern technology was introduced to upgrade Health system readiness for facilitating support for healthcare providers.

Following Technologies were incorporated;

- Real time data entry with notifications enabled for high-risk cases
- On-the-job clinical decision support through integrated clinical rules
- Interactive dashboards, reports, and registers
- 24\*7 specialist support through Remote Support Centre
- Installation of E-partograph
- Games and E-learning content



### Impact generated

ASMAN application was implemented at 42 identified delivery points within one year. More than one lakh cases were recorded, and 40 thousand high risk cases were detected through the application. The initiative also supported LaQshya for monitoring and collecting information about various indicators.

Fig 4 – ASMAN initiative activities



<sup>58</sup> National Health Mission. (2019). *We Care-Good Replicable and Innovative Practices*.

## 2. “Reduction in Malaria Burden Under Intensified Malaria Elimination Project, Meghalaya”<sup>58</sup>



### Background

Large cases of malaria are reported from the North-Eastern States like Tripura and Meghalaya and from the peninsular states of Odisha, Chhattisgarh and Jharkhand. These areas are commonly characterised as hilly, tribal, forested, and conflict-affected areas. Notably, about 80% of the malaria cases are concentrated among the 20% of people classified as “high risk” in these states.

The Global Fund to Fight AIDS, Tuberculosis, and Malaria (GFATM) has been providing support for malaria control and elimination efforts in India since 2005. With the assistance of the Global Fund, progress has been made in accelerating malaria elimination efforts in the states that receive support, bringing them closer to achieving the desired goal of malaria elimination.



### Steps taken by the government

The key strategies of the Global Fund support for malaria control and elimination in India include:

- Early Diagnosis and Complete Treatment (EDCT): Ensuring early diagnosis of malaria cases and providing treatment to affected individuals.
- Integrated Vector Management: Implementing strategies to manage disease vectors effectively, which includes the promotion of Long-Lasting Insecticidal Nets (LLINs) to prevent mosquito bites.
- Intensive Information, Education, and Communication (IEC): Conducting awareness campaigns and education programmes to inform the public about malaria prevention and control measures.
- Capacity Building and Training: Enhancing the skills and knowledge of healthcare workers and community volunteers involved in malaria control efforts.
- Mobility Support: Providing logistical support to facilitate the movement of resources and personnel for effective malaria control activities.
- Surveillance: Establishing and maintaining robust surveillance systems to monitor malaria cases and track transmission patterns.
- Human Resource Support: Ensuring an adequate workforce and support for healthcare professionals and volunteers engaged in malaria control initiatives.



### Impact generated

In 2015, Meghalaya was considered one of the high burden states for malaria in India, contributing to 4.16% of total cases, 5.16% of Plasmodium falciparum (Pf) cases, and 20.58% of malaria-related deaths in the country. However, the state has made significant progress in reducing malaria cases, Pf cases, and deaths. By 2018, there was an impressive 87% reduction in total cases (~6,200), an 86% reduction in Pf cases, and a 92% reduction in deaths compared to the numbers recorded in 2015 (~48,000 total cases). Additionally, in August 2019, there was a 73% reduction in cases, a 75% reduction in Pf cases, and a 100% reduction in deaths when compared to the previous year.

Fig 5 – Training activities for ASHA teams



### 3. “Facilitating quality health services at Health and Wellness Centres in Nagaland”<sup>58</sup>



#### Background

Nagaland faces a low institutional delivery rate, as reported in NFHS-4, only 33% of deliveries have occurred in healthcare facilities. This situation is intensified by inadequate infrastructure at the Sub-Centres, making it unsuitable for conducting deliveries due to factors such as the absence of functional labour rooms, a shortage of trained healthcare personnel, prevailing community preference for home deliveries, and limited transportation options, particularly in the rugged terrain.



#### Steps taken by the government

**Various initiatives were taken to support the healthcare facilities and make the situation better:**

- USAID-MCSP (United States Agency for International Development - Maternal and Child Survival Programme) and Jhpiego have acted as implementation partners for the project.
- Village leaders and other influential figures received training on the importance of community involvement as a means to enhance health outcomes through Health and Wellness Centres (HWCs).
- Village Health Committees (VHCs) assumed responsibility for the enhancement of existing sub-centres into HWCs. These committees ensured transparency in funding and oversaw the construction and upgrading of existing Sub-Centres.
- VHCs played a role in the establishment of a communal kitchen located adjacent to the HWCs, catering to the needs of family members.
- These collective efforts have had a positive impact on the well-being of 82,727 individuals residing in the catchment areas of these 49 HWC-SHC facilities.

Village Health Committees (VHCs) played a vital role in overseeing the construction of wellness rooms at Health and Wellness Centres (HWCs). The majority of the upgrade work was successfully completed with complete transparency, as many local religious leaders and other influential individuals within the village were active members of these health committees. Additionally, the village councils played a crucial role in supervising any development projects within the village.

Among the 49 operationalised Health and Wellness Centres with Sub-Centres (HWC-SHCs), a community kitchen was established through community contributions adjacent to 49 HWC-SHCs. In 19 of these HWC-SHCs, Community Health Officers (CHOs) have initiated institutional deliveries. Patients have been utilising the community kitchen for purposes such as boiling water and cooking food. This has significantly contributed to the promotion of institutional deliveries within the HWC-SHCs. This initiative not only strengthens the Communitisation Act but also fosters greater community ownership in ensuring the provision of healthcare services.

Fig 6 – Health and Wellness Centre in Nagaland



# Recommended Readings

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